

DIEGO RODRILLA MIRHAN

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ABOUT ME

Computer Science student focused on Artificial Intelligence, Machine Learning and full-stack development, taking part in AI research. I build complete AI products: RAG systems with embeddings and vector databases, local LLMs and LLMs through APIs (Anthropic, OpenAI and Gemini), computer vision, OCR and ML models. On the back end I use Python with FastAPI and Node.js, on the front end React with TypeScript, always with Docker, automated tests and CI. I use AI tools in development and publish technical articles on Medium. Advanced English (C1).

WORK EXPERIENCE

Software Developer, *Comando Geek Marketplace* | Jul 2024 – Mar 2025

- Administration, maintenance and feature development on a Geek Marketplace platform, resulting in a 50% increase in user acquisition and retention.
- Performance optimizations and UI/UX improvements using WordPress, PHP and JavaScript. Link: comandogeek.com.br

Family Business | Nov 2019 – Jun 2026

- Customer service (online and in person) — direct communication, teamwork, task and booking management in the hospitality industry, and problem-solving.

RELEVANT PROJECTS

PolyRAG — federated RAG with deterministic routing (2026)

- RAG over SQL, a vector index (Qdrant, FAISS) and a knowledge graph that picks the data source with cosine similarity, with no agent; multilingual BGE-M3 embeddings, FastAPI, 100% local.
github.com/diegormirhan/polyrag

Voice Assistant — 100% local multimodal voice AI (2026)

- Full-duplex voice assistant with screen awareness for Windows: whisper.cpp, llama.cpp and Piper TTS in pt-BR, with real-time interruption, no cloud and no API keys (GGML + Vulkan) — 11 stars on GitHub.
github.com/diegormirhan/voice-assistant

AeroPulse — predictive maintenance for turbofan engines (2026)

- End-to-end ML product using NASA C-MAPSS data: XGBoost for remaining useful life with calibrated uncertainty and explainability, FastAPI API, React 19 + TypeScript dashboard, tests and Docker Compose.
github.com/diegormirhan/aeropulse

Intelligence OCR — local AI document pipeline (2026)

- Asynchronous (asyncio) pipeline that extracts structured data from contracts, invoices and receipts with vision and text LLMs (GLM-OCR, Qwen 2.5 via Ollama), with PostgreSQL/JSONB caching and Docker.
github.com/diegormirhan/intelligence-ocr

Tools4Devs — desktop app with Tauri, Rust and TypeScript (2026)

- All-in-one Windows app to download and convert media, edit images and files, and recognize songs, integrating FFmpeg, yt-dlp, ImageMagick and 22 more tools, with no command line.
github.com/diegormirhan/tools4devs

Other projects (2023–2026)

- LLM Inference Benchmark (4 engines on AMD/ROCm GPU), ProofLens (image authenticity, Playwright E2E), RasterScope (satellite segmentation with U-Net/ONNX), FactoryPulse (failure risk with scikit-learn), ai-memory-win (memory for coding agents, Rust + MCP) and All-in-One Downloader Bot (62 stars). Articles at medium.com/@diegomirhan
github.com/diegormirhan

EDUCATION

Bachelor's Degree — Computer Science, *Universidade Cruzeiro do Sul Virtual (UNIFRAN campus)*

| Feb 2025 – Present

- 4th Semester: Calculus 1 and 2, Linear Algebra, Numerical Calculus and Discrete Mathematics, among others.

TECHNICAL SKILLS

- AI & Machine Learning: LLMs (Anthropic, OpenAI and Gemini APIs; llama.cpp, Ollama, vLLM), RAG, embeddings and vector databases (Qdrant, FAISS), LangChain, LlamaIndex, CrewAI, HuggingFace, PyTorch, XGBoost, scikit-learn, ONNX, computer vision and OCR
- Languages: Python, TypeScript, JavaScript, Node.js, SQL, Rust (basic), PHP
- Back-end & Front-end: FastAPI, Express.js, REST APIs, asyncio, React, Next.js, Tailwind, Streamlit
- Databases: PostgreSQL, MongoDB, Redis, SQLite
- DevOps & Cloud: Docker, Docker Compose, Git, GitHub Actions (CI/CD), Playwright, OpenTelemetry, AWS (EC2, Lambda, CloudWatch, Amplify), Tauri, Electron
- AI in development: Claude Code, Codex, GitHub Copilot, Antigravity
- Spoken languages: Advanced English (C1) — technical reading, speaking and writing; Beginner French (A1)